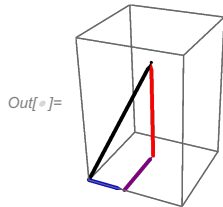


A vector is a list of numbers. In physics, a vector might be a list of position coordinates, as in the Cartesian plane. Descartes wrote about describing the fly in his room using three numbers, one number for the position along one wall, a second for the position along another wall, and a third for how high up the wall.



$$\begin{pmatrix} 1.6 \\ 2.6 \\ 4.3 \end{pmatrix}$$

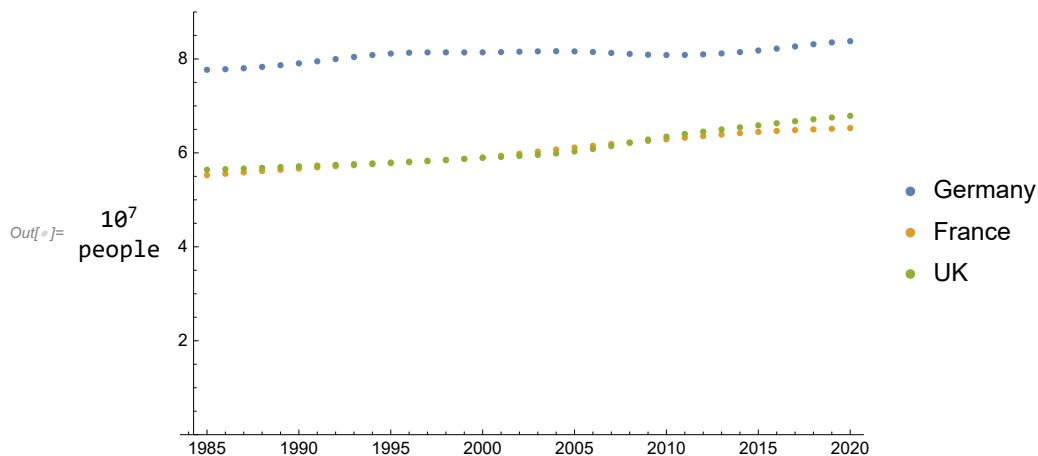
In modern times, numerical data comes in vectors. For example, bivariate data are vectors of length 2. In biology, a vector can be a list of population sizes. For instance, there can be a number of rabbits and a number of foxes in the forest. Or it can be age groups in a population, or the frequencies of different genes.

For example, the height and weight of a penguin is a vector of length 2. If we have a data set of heights and weights, it is a list of vectors, one vector for each penguin. Or there might be more measurements, like body temperature, age, volume, etc.. and then the vectors would be more than length 2.

A vector can be of any length.

If we have a sequence of measurements, where each measurement is itself a list of things, then we get a list of vectors.

Here are the populations of Germany, France, and the UK from 1985 to 2020.



That is just a regular scatter plot, but if we make a table, you can see that each row is a vector.

Out[]:=

Germany	France	UK
7.76916	5.52548	5.64136
7.78061	5.55453	5.65225
7.80157	5.58376	5.66583
7.83075	5.61263	5.68128
7.86603	5.64043	5.69743
7.9054	5.66669	5.71344
7.94908	5.69136	5.72908
7.99632	5.71472	5.7446
8.04286	5.73704	5.76023
8.08329	5.75871	5.77634
8.11387	5.78019	5.79325
8.13237	5.80121	5.81136
8.13993	5.82207	5.83073
8.14027	5.84442	5.851
8.13899	5.87044	5.87159
8.14009	5.90151	5.89233
8.14539	5.93841	5.91243
8.15351	5.98034	5.93263
8.16144	6.02516	5.95614
8.16465	6.0698	5.98728
8.16027	6.11201	6.0288
8.14722	6.15089	6.08213
8.12778	6.18692	6.14552
8.10658	6.22092	6.21451
8.09	6.25429	6.28286
8.0827	6.28795	6.34598
8.08556	6.32222	6.40219
8.09726	6.35642	6.45253
8.11744	6.38935	6.4984
8.14504	6.41935	6.5423
8.17874	6.44532	6.58601
8.21938	6.46676	6.62979
8.26584	6.48425	6.67275
8.31244	6.49905	6.71417
8.3517	6.51297	6.75302
8.37839	6.52735	6.7886

Vectors can be added together.

We do this when we have formulas for vectors, but also when we compute the average of a list of vectors then first we add them all together.

Let's add all of those vectors together which means adding all the rows together to get a single vector.

Out[]:=

Germany	France	UK
291.322	217.155	219.04

Vectors can be multiplied by a number, which in the context of vectors is called a scalar.

We will do this with formulas too, but you also have to divide when you compute the average of a list of vectors.

Let's divide that vector of totals by 36 to get the average.

Out[]=

Germany	France	UK
8.09228	6.03208	6.08445

Vectors can be subtracted, as that is a combination of multiplying by -1 and adding.

Addition is term by term

$$\text{Out[]}= \begin{pmatrix} 1.2 \\ 1.6 \\ 1.5 \end{pmatrix} + \begin{pmatrix} 1.6 \\ 2.6 \\ 4.3 \end{pmatrix} = \begin{pmatrix} 1.2 + 1.6 \\ 1.6 + 2.6 \\ 1.5 + 4.3 \end{pmatrix}$$

$$\text{Out[]}= \begin{pmatrix} 1.2 \\ 1.6 \\ 1.5 \end{pmatrix} + \begin{pmatrix} 1.6 \\ 2.6 \\ 4.3 \end{pmatrix} = \begin{pmatrix} 2.8 \\ 4.2 \\ 5.8 \end{pmatrix}$$

Subtraction is term by term

$$\text{Out[]}= \begin{pmatrix} 1.2 \\ 1.6 \\ 1.5 \end{pmatrix} - \begin{pmatrix} 1.6 \\ 2.6 \\ 4.3 \end{pmatrix} = \begin{pmatrix} 1.2 - 1.6 \\ 1.6 - 2.6 \\ 1.5 - 4.3 \end{pmatrix}$$

$$\text{Out[]}= \begin{pmatrix} 1.2 \\ 1.6 \\ 1.5 \end{pmatrix} - \begin{pmatrix} 1.6 \\ 2.6 \\ 4.3 \end{pmatrix} = \begin{pmatrix} -0.4 \\ -1 \\ -2.8 \end{pmatrix}$$

Multiplication is term by term

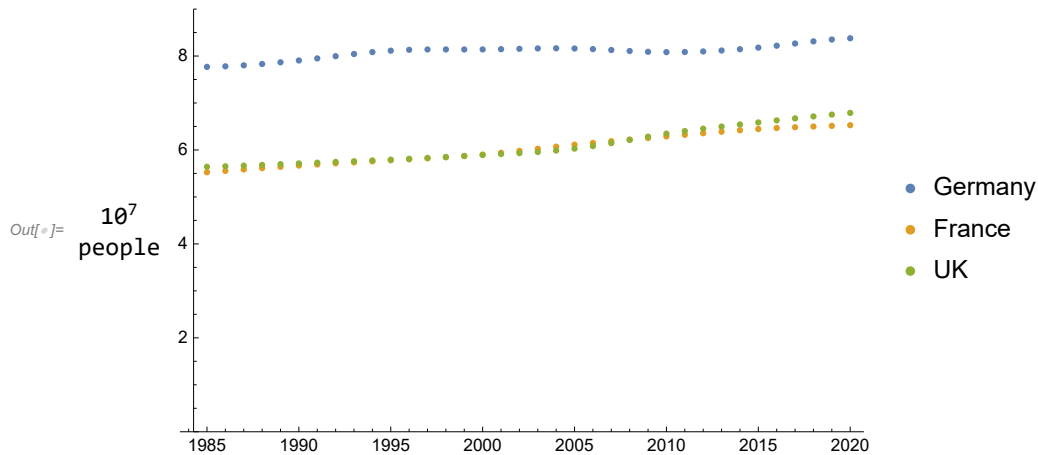
$$\text{Out[]}= 1.6 \cdot \begin{pmatrix} 1.6 \\ 2.6 \\ 4.3 \end{pmatrix} = \begin{pmatrix} 1.6 \times 1.6 \\ 1.6 \times 2.6 \\ 1.6 \times 4.3 \end{pmatrix}$$

$$\text{Out[]}= 1.6 \cdot \begin{pmatrix} 1.6 \\ 2.6 \\ 4.3 \end{pmatrix} = \begin{pmatrix} 2.56 \\ 4.16 \\ 6.88 \end{pmatrix}$$

With these operations, the terms don't interact with each other. Later we will see that matrix multiplication mixes the terms together.

Vector Dynamics

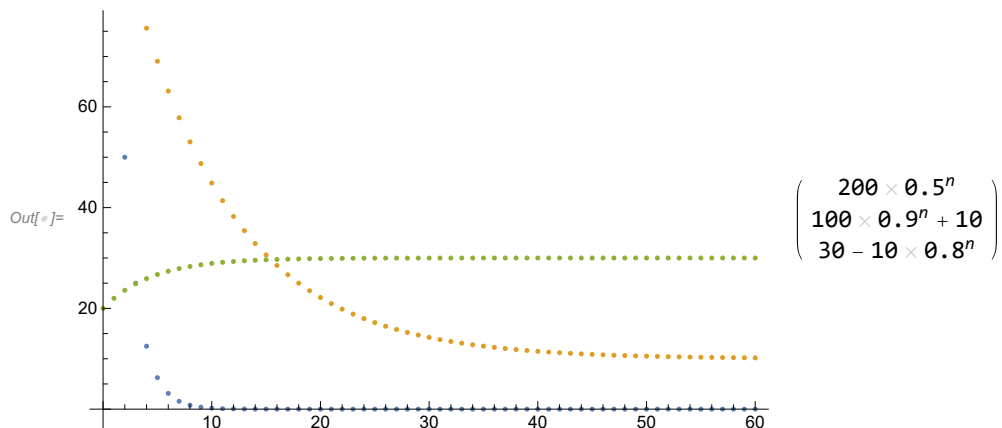
Back to the lists of vectors, we might wonder what govern these vectors. How do they change in time?



Usually with actual measurements they don't follow a recurrence equation to infinite precision.

Instead we make models. This data seems to be roughly linear so we might do a linear regression. But let's turn our attention to other vector dynamics.

What is limiting behavior for vectors? Each term has its own limit. In this case, each term has its own exponential decay.



Recall, that when the base is a constant between 0 and 1 then b^n has the limit equal to 0, so in the formulas you can replace 0.5^n or 0.9^n or 0.8^n with 0 to get the limit.

The limits are 0, 10, and 30 so the vectors converge to the vector

Out[*]= //MatrixForm=

$$\begin{pmatrix} 0 \\ 10 \\ 30 \end{pmatrix}$$

Some terms converge faster than others, but that does not change what it means for the vectors to converge to a limit.

Sequences of numbers can have a recursive rule, a general rule, or both. The same for sequences of

vectors.

Here is the Fibonacci rule for vectors: $v(n+2) = v(n+1) + v(n)$

With the Fibonacci rule starting from the vectors $v(0) = (1, 0, 2, 1)$ and $v(1) = (1, 1, 1, 0)$ the sequence can be continued as in this table.

Out[24]=

1	0	2	1
1	1	1	0
2	1	3	1
3	2	4	1
5	3	7	2
8	5	11	3
13	8	18	5
21	13	29	8
34	21	47	13
55	34	76	21

So far, the components of the vector have not interacted with each other.

Let's consider a tree farm that seels three year old trees. Each year they plant as many new trees as were sold.

The vector will be of the form (baby trees, one-year old trees, two year old trees)

Each year the trees get older. So the number of one year old trees this year is the number of baby trees last year, etc..

Later we will handle this situation with matrices, but for now let's use three variables to make three sequences. Let $x(n)$ be the number of baby trees at time n . Let $y(n)$ be the number of one year old trees at time n . Let $z(n)$ be the number of two year old trees at time n . Then we have two equations from the normal aging of trees:

$$y(n+1) = x(n)$$

$$z(n+1) = y(n)$$

The third equation comes from the assumption that they plant as many new trees as were sold. They seel the oldest trees so

$$x(n+1) = z(n)$$

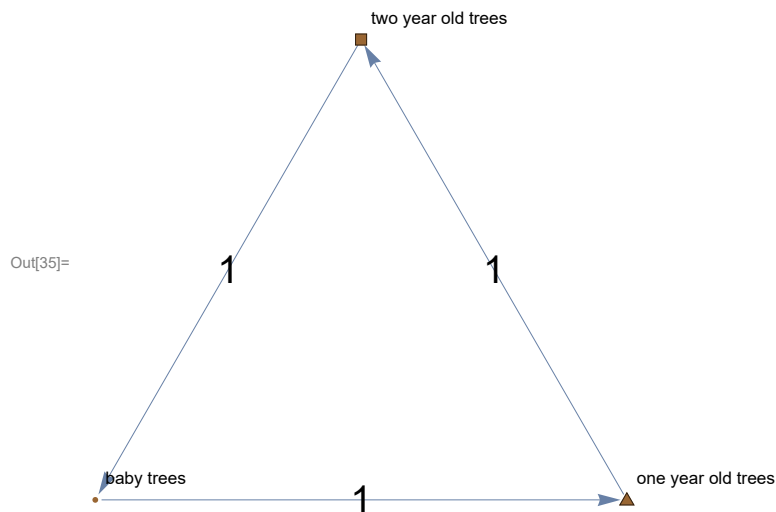
Let's start with the initial vector (20, 50, 40) and this table shows the vector dynamics.

Out[26]=

20	50	40
40	20	50
50	40	20
20	50	40
40	20	50
50	40	20

This one repeats. There is no limit.

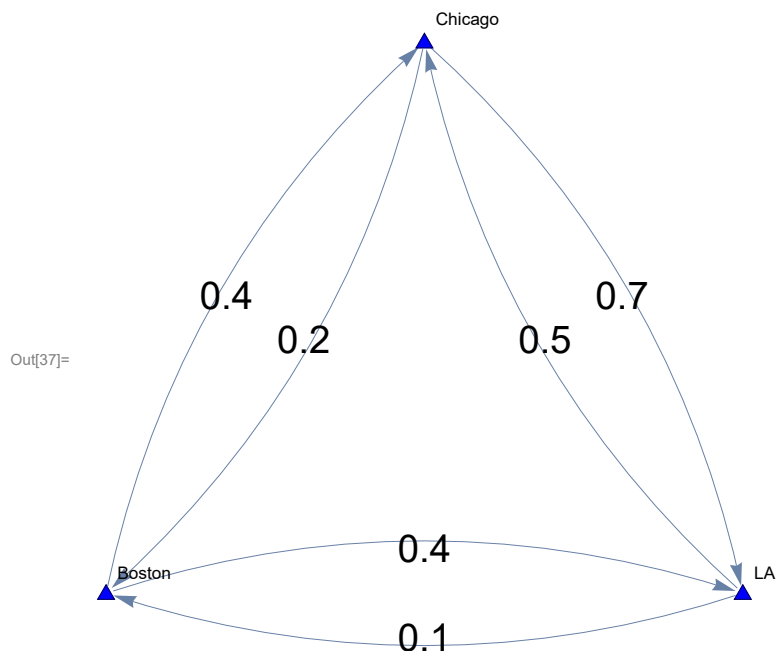
We can use a flow chart for those trees.



The numbers show what fraction goes from one state to another. Normally there is more than one outgoing arrow, and normally the fractions in the outgoing arrows add up to less than one. What remains is the fraction that stays in that state.

Suppose that people move between: Boston, Chicago, LA. At any point in time, we have three numbers. The fraction of the total in Boston, the fraction in Chicago, and the fraction in LA. Let's say at time 0, the initial vector is $(0.5, 0.2, 0.3)$ which means that half of these people are in Boston, 0.2 of them are in Chicago, and 0.3 are in LA.

For this problem this is the flow chart.



Each edge gives the fraction that moves. Let's describe what happens to these people in Boston. The fraction of these people in Boston that move to Chicago is 0.4 and the fraction of these people in

Boston that move to LA is 0.4. What happens to the rest? $1 - (0.4 + 0.4) = 1 - 0.8 = 0.2$. So that means that 0.2 of these people in Boston stay in Boston.

How do we know how many of these people in Boston actually move to Chicago? The fraction is 0.4 and we multiply that by the number of people in Boston. At step 0 that number is 0.5 so the amount in Chicago increases by $(0.4)(0.5) = 0.2$ and the amount in Boston decreases by that same amount.

The initial vector is (0.5, 0.2, 0.3) and now we know how to calculate the second step.

The amount in Boston will be $(0.2)(0.5)$ who remain in Boston + $(0.2)(0.2)$ from Chicago + $(0.1)(0.3)$ from LA = 0.17

The amount in Chicago will be $(0.4)(0.5)$ from Boston + $(0.1)(0.2)$ who remain in Chicago + $(0.5)(0.3)$ from LA = 0.37

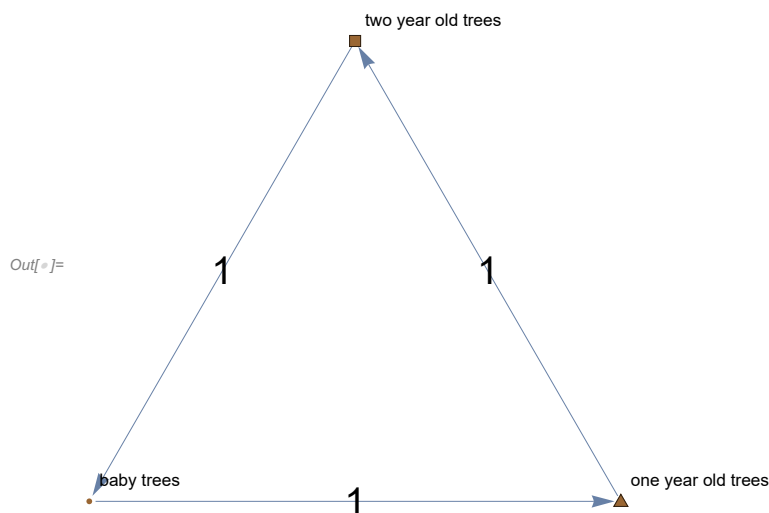
The amount in LA will be $(0.4)(0.5)$ from Boston + $(0.7)(0.2)$ from Chicago + $(0.4)(0.3)$ who remain in LA = 0.46

This seems a lot more complicated. It is simpler to organize this as a matrix. The entry in row i and column j is often denoted with subscripts like a_{ij} and is equal to the fraction from j that move to i .

Out[38]//MatrixForm=

$$\begin{pmatrix} 0.2 & 0.2 & 0.1 \\ 0.4 & 0.1 & 0.5 \\ 0.4 & 0.7 & 0.4 \end{pmatrix}$$

Going back to the trees, we can write that as a matrix too. Wherever an arrow is missing we put a 0.



The matrix for the trees dynamics is

Out[39]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

One thing to notice is that when used in dynamics, the number of rows is the same as the number of columns. In general, matrices don't need to have the rows = columns.

Another thing to notice is that the matrices we have been talking about have another special property.

The total number of people moving around never changed, and the tree farm always had the same total of trees. In these situations the matrix is called a transfer matrix, and its columns add up to 1.

Later we will consider other types of matrices, e.g., Leslie matrices, where the total number of things might change, e.g., the births might outnumber the deaths or vice versa.